

Artem Semidetnov

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Education

BSc	Saint-Petersburg State University , Mathematics	2021 – 2025
BSc	Neapolis University Of Pafos , Applied Computer Science JetBrains Program	2022 – 2024
MSc	Geneva University , Mathematics	2025 – 2027

Work Experience

JetBrains , Research Intern in HoTT and Dependent Types Lab	Remote
- Developed AI features for the Arend language including MCP servers for typechecking, IDE integration. Contributed to the Arend plugin in IntelliJ IDEA. - Developed an experimental LLM coding agent for writing Arend code using framework Koog. - Fine-tuned models for Arend code completion.	Nov 2025 – Apr 2026
JetBrains , Research Intern in HoTT and Dependent Types Lab	Pafos, Cyprus
- I was developing the official library in the Arend language. I formalized different results in algebra and homotopy type theory, including following. - Formalized Eckmann-Hilton argument, Eilenberg-MacLane spaces, Homologies of types. - Formalized automorphisms of groups, Schur's Lemma, Maschke's Lemma, Group actions characterizations. (Some of these results are in the 1.10 release).	June 2024 – Sept 2024
IPONWEB (acquired by Criteo) , Machine Learning Intern	Pafos, Cyprus
Criteo has a ML tool that analyzes sites and produces word2vec representations. Reverse-engineered the behaviour of this tool and analyzed its possible applications.	May 2023 – Sept 2023

Publications

The stable commutator length of a relator is not a one-relator group invariant ↗	Aug 2026
Artem Semidetnov https://doi.org/10.48550/arXiv.2608.21465 ↗	
The operad associated to a crossed simplicial group ↗	Dec 2025
Artem Semidetnov https://doi.org/10.48550/arXiv.2512.14498 ↗	
On the geometry of free nilpotent groups ↗	May 2021
Artem Semidetnov , Ruslan Magdiev https://doi.org/10.48550/arXiv.2106.00095 ↗	

Awards and Scholarships

Scholarship by "Rodnye Goroda" (a social investment program of PJSC "Gazprom neft"), 2023-2024

JetBrains Scholarship in Neapolis University

Travel grant of Chebyshev laboratory for [Winter student school in mathematics and CS of HSE and MCS](#) [↗](#), Moscow

[Excellence Master Fellowship at Geneva University](#) [↗](#)

Teaching Experience

Sirius educational center , Teaching assistant on the course " Groups of intermediate growth " ↗	Sochi, Russia Apr 2024 – May 2024
Laboratory for continuous mathematical education , mathematics teacher for gifted students	Saint-Petersburg, Russia 2021 – 2023
Mathematics and Computer Science faculty program for prospective students , Teaching assistant on the course Braid and knot theory	Saint-Petersburg, Russia June 2023

Selected Talks

On the geometry of free nilpotent groups Siberian summer conference	Novosibirsk, Russia 2021
"On the Poisson boundary of lamplighter groups" ↗ St. Petersburg Seminar on Representation Theory and Dynamical Systems	Saint-Petersburg, Russia
"Twisting numbers on braid and Thompson's groups" ↗ Topological Methods in Dynamics and Related Topics VII	N. Novgorod, Russia 2024
"Plus construction and simplicial methods" ↗ Conference "Algebraic Topology, geometry, combinatorics, and data analysis" ↗	Pushkin, Russia Aug 2025
"The operad associated to a crossed simplicial group" ↗ Young Topologists Meeting 2026	Copenhagen, Denmark July 2026

Miscellaneous

- Finalist of 2020 Intel ISEF
- Intel ISEF alumni
- Winner of 2019 Baltic SEF, PDMI special prize in 2019 Baltic SEF
- 3rd team place in 2019 [International Tournament of Young Mathematicians](#) [↗](#) in Barcelona, Spain
- Organizer of the [Euler International Mathematical Institute's functional analysis seminar](#) [↗](#)
- English level C1 (IELTS 8.0/9, taken in 2020, 2024)
- Invited judge in Saint-Petersburg Tournament of Young Mathematicians (since 2021)
- Created mathematical problem for 2024 International Tournament of Young Mathematicians [10th in here](#) [↗](#)
- Invited judge in International Tournament of Young Mathematicians 2024, 2025
- A member of Scientific Organization Committee in ETEAM 2025 [↗](#)